PROJECT DESCRIPTION REPORT

Hospital Management System

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1. Project Title and Business Idea

The project is titled **Hospital Management System**. It proposes an information system for organizing basic hospital activities involving patients, doctors, appointments, medical records, and bills. The system is represented at the design level using a UML class diagram.

2. Problem Statement

Hospitals need to keep patient information organized and coordinate appointments, medical records, and billing. If these activities are managed separately or manually, information may be difficult to find, appointments may be harder to coordinate, and billing details may be less convenient to track. The proposed system models these key information areas and the relationships between them.

3. Purpose and Objectives

- Organize basic patient information.
- Maintain doctor details, including specialization and contact information.
- Represent appointments with a date, time, and status.
- Keep medical-record information associated with patients.
- Represent bills, total amounts, issue dates, and payment status.
- Show the main classes and their relationships in a UML class diagram.

4. Project Scope

Included in the scope

- Patient information management, including registration and updating details.
- Doctor information and specialization.
- Appointment scheduling, rescheduling, cancellation, and check-in operations.
- Medical-record information, including diagnosis, treatment plan, notes, and creation date.
- Billing information and basic payment processing status.
- A UML class model showing attributes, methods, and relationships.

Excluded from the scope

- Developing and deploying a fully working software application.
- Connecting the system to real hospital databases or external services.
- Implementing advanced functions such as laboratory systems, pharmacy inventory, insurance claims, or inpatient-bed management.
- Using real patient data; the model is for academic design purposes.

5. Main Classes and Their Responsibilities

Patient: Stores patient details such as patient ID, full name, date of birth, gender, and contact information. It supports registration, updating details, viewing appointments, and viewing medical history.

Doctor: Stores doctor details such as doctor ID, full name, specialization, contact information, and salary. It represents operations related to diagnosis, prescriptions, appointment scheduling, and viewing the schedule.

Appointment: Stores appointment ID, date, time, and status. It represents scheduling, rescheduling, cancellation, and check-in.

MedicalRecord: Stores a record ID, diagnosis, treatment plan, notes, and creation date. It represents creating, updating, and viewing a medical record.

Bill: Stores bill ID, total amount, payment status, and issue date. It represents generating a bill, processing payment, and viewing an invoice.

6. Brief System Requirements

- The system should allow patient details to be recorded and updated.
- The system should store doctor information and specialization.
- The system should represent appointments with dates, times, and statuses.
- The system should associate medical records with the relevant patient.
- The system should record billing amounts, issue dates, and payment statuses.
- The design should make the relationships between the five main classes clear.

7. Conclusion

The Hospital Management System provides a clear example of how a hospital-related business idea can be represented through object-oriented analysis. The project identifies five main classes—Patient, Doctor, Appointment, MedicalRecord, and Bill—and describes their responsibilities. The UML class diagram is intended to communicate the proposed structure and relationships of the system; building a fully functional application is not required for this assignment.

Before submission, add the names and student IDs of both group members if required by your lecturer. Review the report so it accurately matches your final diagram.